Xianyu Chen

Ph.D. in Computer Science, University of Minnesota (2025) · Research Scientist – AI, Fasikl Inc.

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RESEARCH INTERESTS

Human-centered multimodal AI: modeling, aligning, and explaining human visual attention; vision-language reasoning; explainable AI; learning from limited data; deep learning for neural signals.

EDUCATION

University of Minnesota, Twin Cities — Minneapolis, MN
Ph.D. in Computer Science (GPA 4.0/4.0) — Sep 2019 – Dec 2025

- Dissertation: *Building Human-Like Machine Intelligence: Advancing Attention by Modeling, Alignment, and Explainability*
- Advisor: Prof. Catherine Qi Zhao. Committee: Profs. Pen-Chung Yew, Yao-Yi Chiang, and Changhyun Choi
- **2026 Distinguished Dissertation Award**, University of Minnesota Graduate School

Sun Yat-sen University — Guangzhou, China
M.E. in Information and Communication Engineering (advisor: Prof. Ming Jiang) — Sep 2015 – Jun 2018
B.E. in Communication Engineering — Sep 2011 – Jun 2015

HONORS AND AWARDS

- **Distinguished Dissertation Award**, University of Minnesota Graduate School
  One of two recipients university-wide; Mathematics, Physical Sciences & Engineering category — 2026
- University of Minnesota nominee, CGS/ProQuest Distinguished Dissertation Award — 2026
- Outstanding Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) — 2026
- Outstanding Reviewer, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) — 2025
- Oral presentation, European Conference on Computer Vision (top 2.3% of submissions) — 2024
- College of Science and Engineering Graduate Fellowship, University of Minnesota — 2019, 2020
- National Scholarship, Ministry of Education of China — 2013, 2017
- Sun Yat-sen University: First Prize, Outstanding Student Scholarship (2013–2017); Outstanding Undergraduate Student Award (2015); Meritorious Winner, Mathematical Contest in Modeling (2013); First Prize, Chinese Mathematics Competitions (2013); Second Prize, National Undergraduate Electronics Design Contest (2013); Second Prize, China Undergraduate Mathematical Contest in Modeling (2014).

EXPERIENCE

Fasikl Inc. (NeuroAI company) — Bloomington, MN
Research Scientist – AI — Oct 2025 – Present
Research Scientist Intern (part-time from Aug 2024) — *May 2024 – Sep 2025*
Research Engineer Intern — *May – Aug 2023*

- Develop deep-learning methods for peripheral neural signals and closed-loop neuromodulation.
- Co-authored SARM [J9], a ConvLSTM that removes electrical-stimulation artifacts from peripheral-nerve recordings, fine-tuned by reinforcement learning on motor-decoding rewards; in benchtop tests it outperformed blanking and linear interpolation.
- Co-inventor on two 2026 patent applications for closed-loop stimulation control and stimulation-parameter optimization [P13, P14].

Visual Information Processing Lab, University of Minnesota — Minneapolis, MN
Ph.D. Researcher (advisor: Prof. Catherine Qi Zhao) — Sep 2019 – May 2024

- *Modeling.* Proposed the first model of human scanpaths in visual question answering, trained with deep reinforcement learning and a loss that separates scanpaths behind correct and incorrect answers; it generalizes to free viewing and visual search [C7].
- *Modeling.* Proposed an individualized scanpath prediction method that personalizes LSTM- and transformer-based scanpath models to each observer; evaluated on four eye-tracking datasets, including one recorded from autistic and neurotypical adults [C10].
- *Alignment.* Co-led VisualHow, a dataset of 20K+ real-life problems with illustrated step-by-step solutions, solution graphs, and human attention annotations [C9]; then built a graph-attention model whose attention, supervised by these annotations, improves step-by-step solution prediction [J7].
- *Explainability.* Proposed GazeXplain, which predicts a scanpath and describes in natural language what each fixation looks at, and built a LLaVA-assisted, human-verified pipeline to annotate such descriptions on three eye-tracking datasets [C11].
- *Learning from limited data.* Developed human-attention-guided novel object captioning [C8], ensemble self-distillation for few-shot image captioning [C6], and bottom-up/top-down attention for few-shot object detection [R1].
- Co-inventor on a University of Minnesota patent application for rating essential-tremor severity from hand-drawn spirals with deep learning [P12].

Tencent AI Lab · *Research Intern* — Summer 2019

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences — Shenzhen, China
Research Intern, Multimedia Research Center (supervisors: Profs. Yali Wang and Yu Qiao) — Aug – Dec 2018

- Designed DID, which combines architecture disentangling, weight imprinting, and distillation for continuous low-shot object detection [J6, P8]; contributed to Context-Transformer for few-shot detection [C5].

Wireless Communication Lab, Sun Yat-sen University — Guangzhou, China
Student Researcher (advisor: Prof. Ming Jiang) — Dec 2013 – Jun 2018

- Signal processing for wireless and visible light communication (VLC): Bayesian MMSE channel estimation for VLC with performance bounds [J1, C1, P1], adaptive polar-linear interpolation for pilot-aided channel estimation [J5, C4, P2], and optimal dimming-control waveform design for VLC [P9, P10].

PUBLICATIONS

Xianyu Chen in bold; *equal contribution. Ming Jiang on [J7], [C6]–[C11], [R1] and [P12] is Dr. Ming Jiang (University of Minnesota), not Prof. Ming Jiang (Sun Yat-sen University). Online: chenxy99.github.io/publications.html.

Doctoral Dissertation

[D1] **Xianyu Chen**. Building Human-Like Machine Intelligence: Advancing Attention by Modeling, Alignment, and Explainability. Ph.D. dissertation, University of Minnesota, Nov. 2025; xvi + 163 pp. Chapters 2–6 appeared as [C7], [C10], [C9], [J7], and [C11]. [full text]

Conference Papers

[C11] **Xianyu Chen**, Ming Jiang, and Qi Zhao. GazeXplain: Learning to Predict Natural Language Explanations of Visual Scanpaths. In *European Conference on Computer Vision (ECCV)*, LNCS vol. 15066, pp. 314–333, 2024. **Oral presentation.** [doi] [arXiv] [code] [data] [video]

[C10] **Xianyu Chen**, Ming Jiang, and Qi Zhao. Beyond Average: Individualized Visual Scanpath Prediction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25420–25431, 2024. [doi] [paper] [code] [video]

[C9] Jinhui Yang*, **Xianyu Chen***, Ming Jiang, Shi Chen, Louis Wang, and Qi Zhao. VisualHow: Multimodal Problem Solving. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15606–15616, 2022. [doi] [paper] [project] [data]

[C8] **Xianyu Chen**, Ming Jiang, and Qi Zhao. Leveraging Human Attention in Novel Object Captioning. In *International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 622–628, 2021. [doi] [code]

[C7] **Xianyu Chen**, Ming Jiang, and Qi Zhao. Predicting Human Scanpaths in Visual Question Answering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10871–10880, 2021. [doi] [paper] [code] [video]

[C6] **Xianyu Chen**, Ming Jiang, and Qi Zhao. Self-Distillation for Few-Shot Image Captioning. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 545–555, 2021. [doi] [code]

[C5] Ze Yang, Yali Wang, **Xianyu Chen**, Jianzhuang Liu, and Yu Qiao. Context-Transformer: Tackling Object Confusion for Few-Shot Detection. In *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 34, no. 7, pp. 12653–12660, 2020. [doi] [arXiv]

[C4] **Xianyu Chen** and Ming Jiang. Low-Complexity Adaptive Channel Estimation. In *IEEE 88th Vehicular Technology Conference (VTC2018-Fall)*, 2018. [doi]

[C3] Yufa Chen, Ming Jiang, Lin Zhang, and **Xianyu Chen**. Polarity Modulated Complex Colour Shift Keying for OFDM-Based Visible Light Communication. In *IEEE/CIC International Conference on Communications in China (ICCC)*, 2017. [doi]

[C2] Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen**, and Weikun Hou. Miller-Coded Asynchronous Visible Light Positioning System for Smart Phones. In *IEEE 85th Vehicular Technology Conference (VTC2017-Spring)*, 2017. [doi]

[C1] **Xianyu Chen** and Ming Jiang. Enhanced Bayesian MMSE Channel Estimation for Visible Light Communication. In *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2016. [doi]

Journal Articles

[J9] Brian Z.H. Lim, Markus Drealan, Anh Tuan Nguyen, Byeongchan Jeong, Maral Daneshyan, **Xianyu Chen**, Qi Zhao, and Zhi Yang. SARM: Benchtop Evaluation of a Task-Driven Two-Stage Deep Learning Framework for Stimulation Artifact Removal in Peripheral Neural Decoding. *IEEE Transactions on Biomedical Engineering (TBME)*, early access, Sep. 2026. [doi]

[J8] Yaohui Guo, Ling Xu, **Xianyu Chen**, and Zifeng Zhao. A Two-Stage Feature Reduction (FIRRE) Framework for Improving Artificial Neural Network Predictions in Civil Engineering Applications. *Infrastructures*, vol. 11, no. 1, Art. no. 29, Jan. 2026. [doi]

[J7] **Xianyu Chen**, Jinhui Yang, Shi Chen, Louis Wang, Ming Jiang, and Qi Zhao. Every Problem, Every Step, All in Focus: Learning to Solve Vision-Language Problems With Integrated Attention. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 46, no. 7, pp. 4720–4735, Jul. 2024. [doi] [code] [data]

[J6] **Xianyu Chen***, Yali Wang*, Jianzhuang Liu, and Yu Qiao. DID: Disentangling-Imprinting-Distilling for Continuous Low-Shot Detection. *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 7765–7778, 2020. [doi] [code]

[J5] **Xianyu Chen** and Ming Jiang. Enhanced Adaptive Polar-Linear Interpolation Aided Channel Estimation. *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 3, pp. 693–696, Jun. 2019. [doi]

[J4] Huaiyin Lu, Lin Zhang, **Xianyu Chen**, and Zhiqiang Wu. Recursive Carrier Interferometry Aided High Data Rate OFDM Systems With PAPR Suppression, Phase Noise Rejection, and Carrier Frequency Offsets Compensation. *IEEE Transactions on Vehicular Technology (TVT)*, vol. 68, no. 4, pp. 3655–3671, Apr. 2019. [doi]

[J3] Kuan Wu, Ming Jiang, Fengxia She, and **Xianyu Chen**. Relay-Aided Request-Aware Distributed Packet Caching for Device-to-Device Communication. *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 1, pp. 217–220, Feb. 2019. [doi]

[J2] Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen**, and Weikun Hou. Space-Time-Multiplexed Multi-Image Visible Light Positioning System Exploiting Pseudo-Miller-Coding for Smart Phones. *IEEE Transactions on Wireless Communications (TWC)*, vol. 16, no. 12, pp. 8261–8274, Dec. 2017. [doi]

[J1] **Xianyu Chen** and Ming Jiang. Adaptive Statistical Bayesian MMSE Channel Estimation for Visible Light Communication. *IEEE Transactions on Signal Processing (TSP)*, vol. 65, no. 5, pp. 1287–1299, Mar. 2017. [doi]

Preprint

[R1] **Xianyu Chen**, Ming Jiang, and Qi Zhao. Leveraging Bottom-Up and Top-Down Attention for Few-Shot Object Detection. *arXiv:2007.12104*, 2020. [arXiv] [code]

PATENTS

11 granted patents and 3 pending applications. English titles of Chinese filings are translations.

Pending applications (2026)

[P14] Jules Anh Tuan Nguyen and **Xianyu Chen**. Intelligent Stimulation Control Method Based on Multidimensional Neural and Electromyographic Signals. Chinese patent application, Pub. No. CN122272998A, filed Apr. 2026.

[P13] Jules Anh Tuan Nguyen and **Xianyu Chen**. Dynamic Stimulation Parameter Optimization Method and System in a Closed-Loop Neuromodulation System. Chinese patent application, Pub. No. CN122399237A, filed Mar. 2026.

University of Minnesota application (2023)

[P12] Qi Zhao, Ming Jiang, **Xianyu Chen**, and Zhi Yang. Method, Apparatus and System for Recognizing Tremor Symptom, Recognition Terminal and Storage Medium. U.S. Patent Application Pub. No. US 2024/0032819 A1 (Appl. No. 18/448,995, filed Aug. 2023; assigned to the University of Minnesota); Chinese counterpart CN118196594A.

Granted patents (China)

[P11] Ming Jiang, Junyu Chen, Lei Zhao, and **Xianyu Chen**. Channel Estimation Method of Marine Communication System. CN111404847B, filed 2020, granted 2021.

[P10] Ming Jiang and **Xianyu Chen**. Dimming Control Visible Light Communication System and Method Based on Convex Optimization Framework. CN110855364B, filed 2019, granted 2023.

[P9] Ming Jiang and **Xianyu Chen**. Optimal Dimming Control Visible Light Communication System and Method Based on Distortion Strategy. CN110855363B, filed 2019, granted 2022.

[P8] Yu Qiao, **Xianyu Chen**, and Yali Wang. Target Detection Method, Device and Equipment for Continuous Small Sample Images. CN110033026B, filed 2019, granted 2021.

[P7] Ming Jiang, Zhengpeng Li, Kunyi Cai, and **Xianyu Chen**. Dual-Mode LED Positioning Method and System Based on Different Lambertian Radiation Lobe Moduli. CN109324311B, filed 2018, granted 2022.

[P6] Ming Jiang, Kuan Wu, Fengxia She, and **Xianyu Chen**. Relay D2D Data Packet Caching Method Based on User Grouping Demand Diversity. CN108668261B, filed 2018, granted 2020.

[P5] Ming Jiang, Zefeng Lin, and **Xianyu Chen**. OFDM System Supporting Color Adjustment and CSK Constellation Diagram Detection. CN108092714B, filed 2017, granted 2021.

[P4] Ming Jiang, Jianhui Li, **Xianyu Chen**, Yixue Lei, and Yunfei Zhang. Internet-of-Vehicles Human-Vehicle Resource Allocation Method Based on Geographic Region Information. CN107659915B, filed 2017, granted 2020.

[P3] Ming Jiang, Yufa Chen, Lin Zhang, and **Xianyu Chen**. Optical OFDM Communication System Based on Polar Modulation and Complex Color Shift Keying. CN107395278B, filed 2017, granted 2020.

[P2] Ming Jiang and **Xianyu Chen**. Channel Estimation Method Based on Enhanced Self-Adaptive Polarization Linear Interpolation. CN107508777B, filed 2017, granted 2020.

[P1] **Xianyu Chen** and Ming Jiang. Visible Light Channel Estimation Method and System. CN105471777B, filed 2015, granted 2019.

DATASETS

- **GazeXplain** [C11]: LLaVA-assisted, human-verified natural-language explanations for 86K+ scanpaths in three eye-tracking datasets. [data]
- **VisualHow** [C9, J7]: 20K+ real-life problems with illustrated step-by-step solutions, solution graphs and human attention annotations. [project] [data] [SGAN data]

TALKS AND PRESENTATIONS

- ECCV 2024, Milan, Italy (oral): "GazeXplain: Learning to Predict Natural Language Explanations of Visual Scanpaths" Oct 2024
- IJCAI 2021 (virtual): "Leveraging Human Attention in Novel Object Captioning" Aug 2021
- WACV 2021 (virtual): "Self-Distillation for Few-Shot Image Captioning" Jan 2021

TEACHING EXPERIENCE

Teaching Assistant, Department of Computer Science and Engineering, University of Minnesota

- CSCI 5521: Machine Learning Fundamentals Fall 2021, Fall 2022, Fall 2023, Spring 2024
- CSCI 2033: Linear Algebra Spring 2023
- CSCI 5302: Analysis of Numerical Algorithms Spring 2022

TECHNICAL SKILLS

Methods: computer vision, vision-language models, eye-tracking (gaze) modeling, explainable AI, reinforcement learning fine-tuning, transformers, LSTMs, graph neural networks, knowledge distillation, few-shot learning, biosignal time series, statistical signal processing.

Tools: Python, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, OpenCV, C/C++, MATLAB, Linux, Git.

PROFESSIONAL SERVICE

Journal reviewer: IEEE TPAMI (2023–), IEEE TIP (2026–), IEEE TNNLS (2023–), IEEE TMM (2023–), IEEE TCSVT (2025–), ACM TOMM (2025–), Journal of Visual Communication and Image Representation (2021–).

Conference reviewer: CVPR (2022, 2023, 2025, 2026), ICCV (2023, 2025), ECCV (2022, 2024, 2026), AAAI (2023–2027), WACV (2022–2026), BMVC (2026).